Roll No. .....

Total Page No. : 4

31N0404, 31N0904, 31N1404, 31N1504, 31N1604, 31N1704, 31N1804, 31N1904, 31N2004, 31N2104, 31N2304, 31N2404

B. TECH. III Sem Main/ Back Exam

Academic Session 2024-2025

Computer Science and Engineering and Allied Branches

(3CS4 - 04) - Object Oriented Programming Using C++

Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours] **[Max. Marks : 70**

[Min. Passing Marks :

Instructions to Candidates :

Part-A : Short answer questions (up to 25 words) 10 * 2 marks = 20marks. All ten questions are compulsory.

Part-B : Analytical/Problem Solving questions 5 * 4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part-C : Descriptive/Analytical/Problem Solving questions 3 * 10 marks = 30 marks. Candidates have to answer 3 questions out of 5. Schematic diagrams must be shown wherever necessary. Any data

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you feel missing may suitably be assumed and stated clearly.
Units of quantities used/calculated must be stated clearly.
Use of following supporting materials is permitted during examination.
(Mentioned in form No. 205)

1_____ 2_____

10x2=20

Part-A

Q.1 Explain concept of abstract class. [2]

Q.2 State the use of new operator. [2]

Q.3 What is the role of classes in C++ ? [2]

Q.4 What is Inline Function ? [2]

Q.5 What is the application of : : operator ? [2]

Q.6 What is Virtual base class ? [2]

Q.7 What is late binding ? [2]

Q.8 What is the use of anonymous namespace ? [2]

Q.9 Why do we need Virtual destructor ? [2]

Q.10 Define this pointer. [2]

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Part-B

5x4=20

Q.1 What are the various benefits and application of object oriented programming ? [4]

Q.2 What is operator overlanding ? What is it's need ? Explain with example. [4]

Q.3 What are base and derived classes ? How do they help in interitance ? [4]

Q.4 What is Static Variable ? Why is it called class variable ? Write a program to illustrate its application. [4]

Q.5 Write a program to compare the two given strings using a pointer. [4]

Q.6 What is an exception ? Why it is necesary to handle an exception by Special Code ? Illustrate with an example. [4]

Q.7 What is dynamic memory allocation ? Explain with the help of an example, how to create and destroy objects dynamically. [4]

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Part-C

10x3=30

Q.1 Discuss the term function overloading.

Write a program using function overloading for subtracting two given integer matrices; two floating matrices and double precision matrices separately. [10]

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Q.2 How multi level Inheritance is different from multiple inheritance ? What ambiguity arises in multiple inheritance and how it is resolved ? [10]

Q.3 Define Operator Overloading. Explain how to overload unary operator and binary operator. [10]

Q.4 What is the use of default copy constructor ? Is a constructor and destructor mandatory for a class. Explain by giving examples in each case. [10]

Q.5 What is a Template ? Explain with the help of an example how to creat a Function Template and Class Template. [10

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